

INTEGRATIVE PHYSIOLOGY (POST LEC)

Lecture #1: CARDIAC MUSCLE | OPHT 1-1 | MIDTERMS

By Accel Jimenez

THREE MAJOR TYPES OF CARDIAC MUSCLE

1. Atrial Muscle
2. Ventricular Muscle
3. Specialized Excitatory and conductive muscle fibers

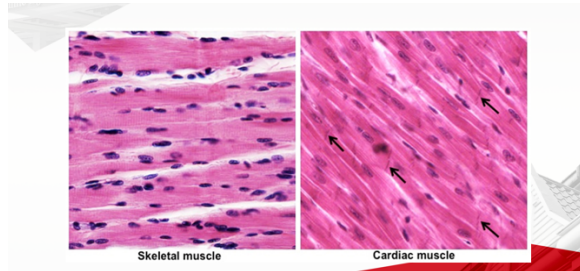


Figure 1

Cardiac Output (CO)

- Amount of blood ejected by the heart per unit of time

Venous Return (VR)

- Amount of blood returning back to the heart per unit of time

Central Venous Pressure (CVP)

- Equal to Right Atrial pressure (RAP). Reflects the amount of blood that returns to the heart.

Peripheral Venous Pressure

- Pressure in the peripheral veins that drain the body tissues.

IMPORTANT PRINCIPLES

- **Cardiac Output (CO)**
 - $CO = HR \times SV = VR$
 - CO LEFT (L) HEART + CO RIGHT (R) HEART
 - CO L heart: Systemic blood flow
 - CO R heart: Pulmonary blood flow
- At rest: 5L/min
- Max CO (non-athlete): 20L/min
- Max CO (athlete): 30L/min
- **Pulmonic Circulation**, compared to systemic Circulation, has:
 - Lower resistance, afterload, stroke work
 - Same preload, heart rate, stroke volume, cardiac output.

- **Stroke Volume (SV)**

- Blood Ejected by the ventricle per heartbeat
- Equal to End Diastolic Volume (EDV) - End Systolic Volume
- Normal value: 70 mL

- **Ejection Fraction (EF)**

- Percentage of EDV that is actually ejected by the ventricle
- Equal to SV/EDV
- Normal value: 55%

- **Cardiac Output (CO)**

- Total blood volume ejected per unit of time
- Equal to $HR \times SV$
- Normal value: 5L/min

- **Stroke Work**

- Work heart performs on each beat.

- **Systole**

- Where the heart chambers are contracting and pumps the blood towards the periphery via the arteries

- **Diastole**

- Where the heart chamber is in a state of relaxation and fills with blood that receives from the veins.

CARDIAC MUSCLE ACTION POTENTIAL

- **Phase 0 (depolarization)**

- Fast sodium channels are open
- Membrane potential becomes more positive.

- **Phase 1 (Initial repolarization)**

- Fast sodium channels close
- Leave the cell through open potassium channels

- **Phase 2 (Plateau)**

- Calcium channels open and fast potassium channels close
- Combination of decreased potassium ion efflux and increased calcium ion influx causes the action potential to plateau

- **Phase 3 (Rapid repolarization)**

- Calcium channels close and slow potassium channels open

- **Phase 4 (Resting membrane potential)**

- Average about -90 millivolts

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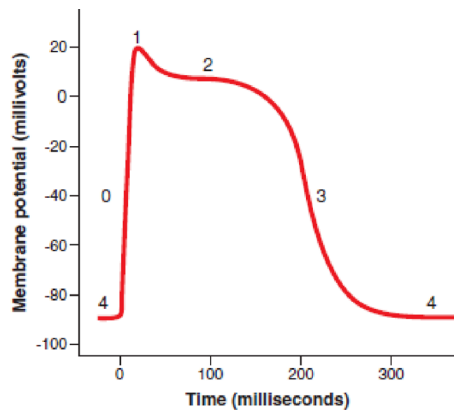


Figure 2

CARDIAC CYCLE

Occur from the beginning of one heartbeat to the beginning of the next

- **Sinus node**
- **Period of relaxation** called **diastole**, during which the heart fills with blood, followed by a period of contraction called **systole**

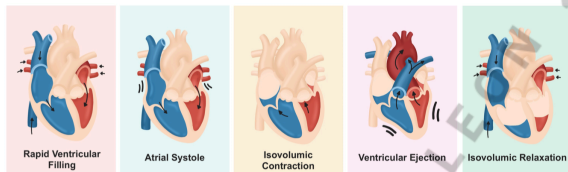


Figure 3

Refractory period of Cardiac Muscle

- The normal refractory period of the ventricle is 0.25 to 0.30 second, which is about the duration of the prolonged plateau action potential
- Atria > Ventricles

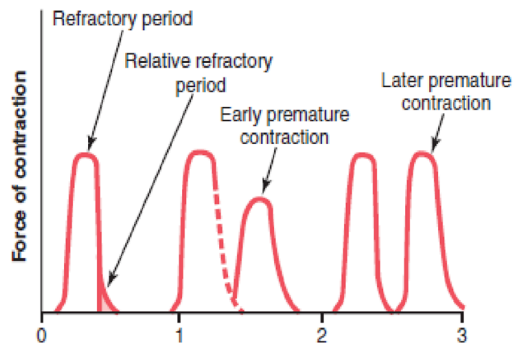


Figure 4

EXCITATION-CONTRACTION COUPLING

- Refers to the mechanism by which the action potential causes the myofibrils of muscle to contract.
- Transverse T tubules → longitudinal sarcoplasmic tubules to cause release of calcium ions → diffuse into the myofibrils → promote sliding of the actin and myosin filaments → muscle contraction.

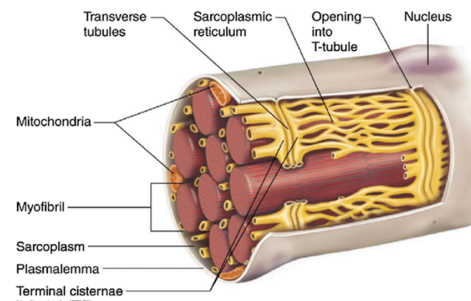


Figure 5

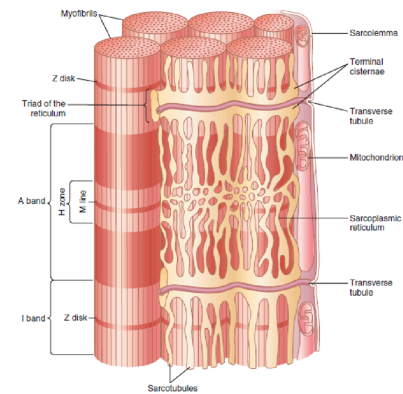


Figure 6

INCREASING HEART RATE DECREASES DURATION OF CARDIAC CYCLE

- When heart rate increases, the duration of each cardiac cycle decreases, including the contraction and relaxation phases.

*refer to figure fig. 3

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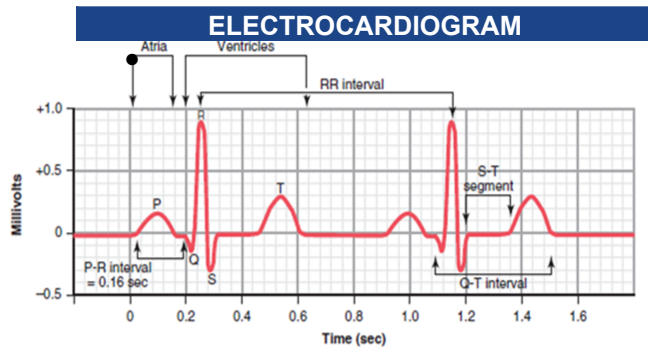


Figure 7

- **P Wave:** caused by spread of **depolarization**
- **QRS waves:** appear as result of **electrical depolarization of the ventricles**
- **Ventricular T wave:** represent the stage of **repolarization of the ventricles.**

DESCRIPTION	TERM
Atrial Depolarization	P WAVE
Ventricular Depolarization	QRS COMPLEX
Ventricular Repolarization	T COMPLEX
Occurs at the start of atrial depolarization to start of ventricular depolarization in the ECG. During this time, cardiac AP passes through the AV node	PR INTERVAL
Period of Ventricular Depolarization + Ventricular Repolarization	QT INTERVAL
AV Node Conduction	PR SEGMENT
Isoelectric portion of the ECG: correlates with Phase 2 (plateau phase) of Ventricular Action Potential	ST SEGMENT

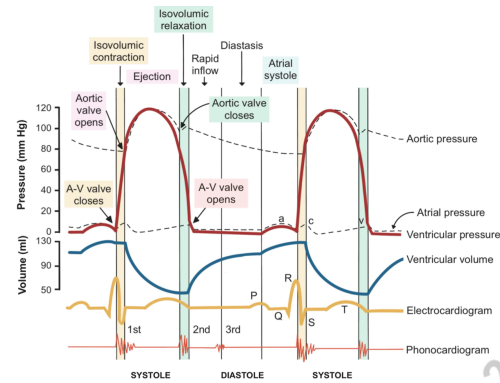


Figure 7

7 PHASES OF CARDIAC CYCLE

1. Atrial Contraction/Atrial Systole
2. Isovolumic Contraction
3. Rapid Ventricular Ejection
4. Slow/Reduced Ventricular Ejection
5. Isovolumic Relaxation
6. Rapid Ventricular Filling (occurs during early third of diastole)
7. Slow/Reduced Ventricular Filling (occurs during middle third of diastole)

1. ATRIAL CONTRACTION

- Occurs during the **distal third of diastole**
- **NOT essential for ventricular filling**
- ECG: **preceded by p-wave**
- **Atrial pressure:** increases slightly
- **Ventricular pressure:** increases slightly
- **Ventricular volume:** increases slightly

2. ISOVOLUMIC CONTRACTION

- ECG: **preceded by QRS complex**
- **Ventricular pressure:** increases but ventricular pressure is still less than aortic pressure
- Semilunar valves are still closed
- Blood will **NOT** flow from LV to Aorta
- **Ventricular volume:** remains the same
- Ventricular pressure > Atrial pressure
- AV valves will close
- **Heart Sounds:** S1 will be heard

3. VENTRICULAR EJECTION

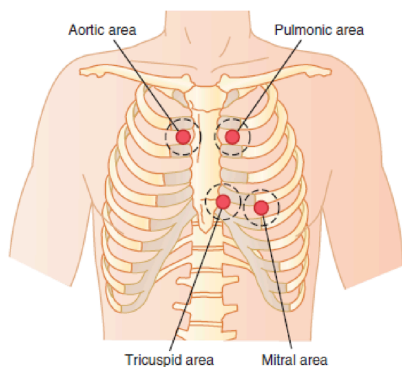
- **Atrial fulling begins**
- **Ventricular pressure:** rapidly increases to a point that it is now greater than **Aortic pressure**
 - Semilunar valves open: blood will flow from LV to aorta
- **Ventricular volume:** rapidly decreases

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- During ventricular ejection, pressure difference is smallest in magnitude between Left Ventricle and Aorta
- 4. REDUCED VENTRICULAR EJECTION**
 - ECG: **T-wave occurs**
 - **Ventricular pressure:** decreases
 - **Ventricular volume:** decreases
 - Aortic pressure decreases because of runoff of blood from large arteries to smaller arteries
- 5. ISOVOLUMIC RELAXATION**
 - ECG: **preceded by T-wave**
 - Aortic valve closes during this portion of the ECG
 - **Ventricular pressure:** rapidly decreases but Ventricular pressure > Atrial pressure
 - **Ventricular volume:** remains the same
 - AV valves are still closed: no blood flow from Atria to Ventricles
 - Ventricular pressure < Aortic pressure
 - Semilunar valves will close
 - **Heart sounds:** S2 is heard
- 6. RAPID VENTRICULAR FILLING**
 - **Ventricular pressure:** rapidly decreases to a point that it is now less than Atrial pressure
 - **Opening of the AV valves:** blood rapidly flows from Atrium to Ventricles
 - **Heart sounds:** 3rd heart sound may be heard (due to rapid ventricular filling)
 - Ventricular volume: rapidly increases.
- 7. REDUCED VENTRICULAR FILLING**
 - **Longest phase of the cardiac cycle**
 - Dependent on heart rate
 - **Ventricular volume:** reduced increase



HEART SOUNDS

- S1**
 - Longer, louder, lower in pitch
 - Closure of AV valves
- S2**
 - Shorter, not louder than S1
 - Closure of SL valves
- S3**
 - **Rapid filling of ventricles**
 - Ventricular gallop (pathological)
 - Congenital heart failure
- S4**
 - From atrial systole
 - Myocardial infarction, hypertension
 - Atrial Gallop (pathological)

HEART SOUNDS

HEART SOUND	CAUSE	PHASE
S1	• Closure of AV Valves	• Isovolumic Contraction
S2 (split during inspiration)	• Closure of Semilunar Valves	• Isovolumic Relaxation
S3	• Rapid Ventricular Filling • (Normal in children, young adults, pregnant patients; • L-Sided S3 in patients with CHF indicative of CV morbidity, mortality)	• Rapid Ventricular Filling
S4	• Stiff Ventricles / Vibration in the ventricular wall during systole	• Atrial Contraction/ Systole

PRELOAD VS AFTERLOAD

- The degree of tension on the muscle when it begins to contract, which is called the **preload**
- The load against which the muscle exerts its contractile force, which is called the **afterload**

REGULATION OF HEART PUMPS

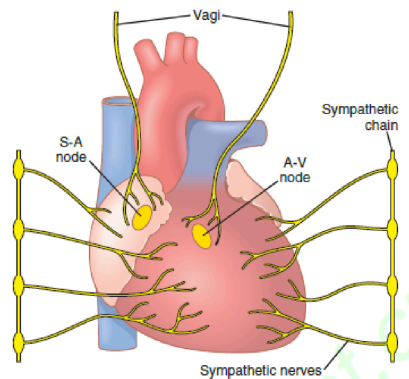
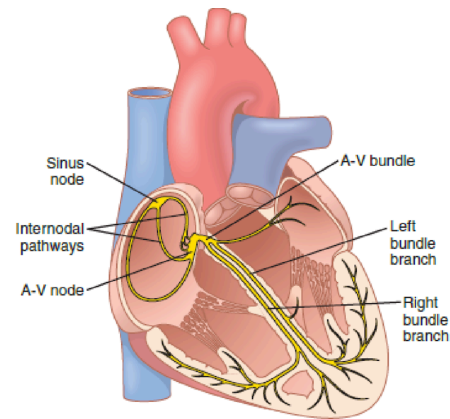
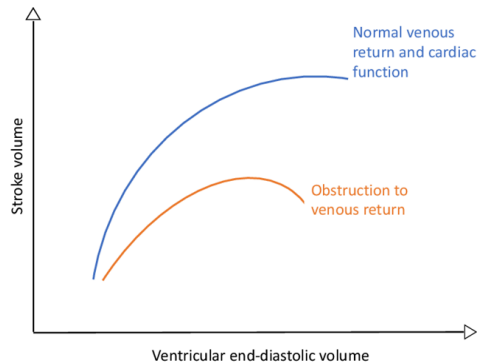
INTRINSIC REGULATION OF HEART PUMPING—THE FRANK-STARLING MECHANISM

- The greater the heart muscle is stretched during filling, the greater is the force of contraction and the greater the quantity of blood pumped into the aorta

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EFFECT OF POTASSIUM IONS

- Makes the heart dilated, flaccid
- Slows the HR
- Weakness of the heart, abnormal rhythm, and death

EFFECTS OF CALCIUM IONS

- Spastic contraction
- Cardiac weakness

EFFECT OF TEMPERATURE

- Increases HR = fever

CHAPTER 10

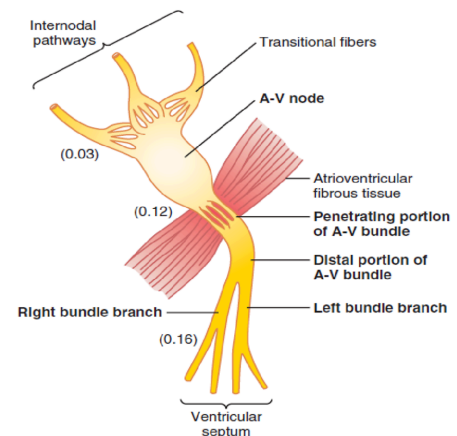
RHYTHMICAL EXCITATION OF THE HEART

SINUS (SINOATRIAL NODE)

- A small, flattened, ellipsoid strip of specialized cardiac muscle
- Located in the superior posterolateral wall of the right atrium immediately and slightly lateral to the opening of the superior vena cava
- (-) contractile muscle

AUTOMATIC ELECTRICAL RHYTHMICALITY OF SINUS FIBERS

- **Self Excitation**
 - A process that cause automatic rhythmical discharge and contraction.
- **Influx of Sodium and Calcium**



AV NODE DELAYS

- Atrial conductive system is organized so that the cardiac impulse does not travel from the atria into the ventricles too rapidly
- This delay allows time for the atria to empty their blood into the ventricles before ventricular contraction begins

AV NODE

- A-V node is located in the posterior wall of the right atrium immediately behind the tricuspid valve

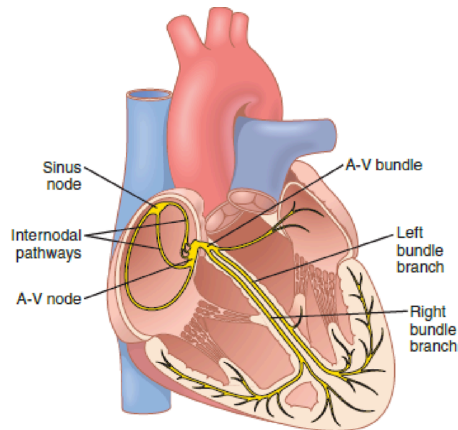
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PURKINJE FIBERS

- Rapid transmission compared to AV nodal fibers



PARASYMPATHETIC STIMULATION OF THE HEART

- **Cranial nerve responsible:** Vagus Nerve
- **Neurotransmitter responsible:** Acetylcholine

SYMPATHETIC STIMULATION OF THE HEART

- increases the rate of sinus nodal discharge.
- increases the rate of conduction, as well as the level of excitability in all portions of the heart
- increases greatly the force of contraction of all the cardiac musculature, both atrial and ventricular
- **Hormone Responsible:** **NOREPINEPHRINE** -> turn stimulates beta-1 adrenergic receptors, which mediate the effects on heart rate

ANS EFFECT ON THE HEART AND BLOOD VESSELS

	SYMPATHETIC		PARASYMPATHETIC	
	Effect	Receptor	Effect	Receptor
Heart Rate	↑	β_1	↓	M_2
Conduction Velocity (AV Node)	↑	β_1	↓	M_2
Contractility	↑	β_1	↓ (atria only)	M_2
VASCULAR SMOOTH MUSCLE				
Skin, splanchnic	Constriction	α_1	Dilation (EDRF)	M_3
Skeletal Muscle	Constriction	α_1	Dilation (EDRF)	M_3
	Dilation	β_2	-	-
Veins	Constriction	α_1	-	-